8 **(a)** Define *electric field strength* at a point.

.....

.....

..... [1]

(b) Fig. 8.1 shows two horizontal plates A and B whose electric potentials are 70.0 V and –20.0 V respectively. The plates are 90.0 cm apart. A conductor P is situated centrally between the two plates.

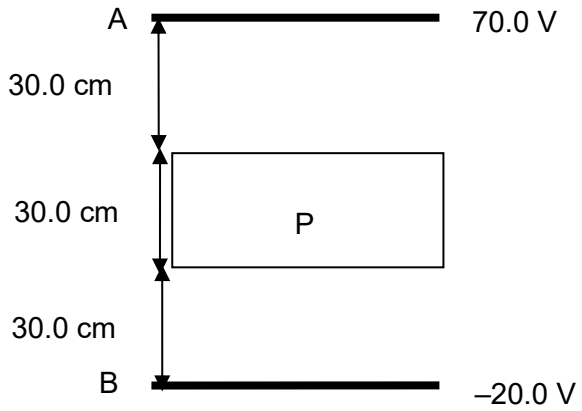

Fig. 8.1

(i) On Fig. 8.2, sketch the variation of electric potential *V* with distance *x* from A, between the two plates.

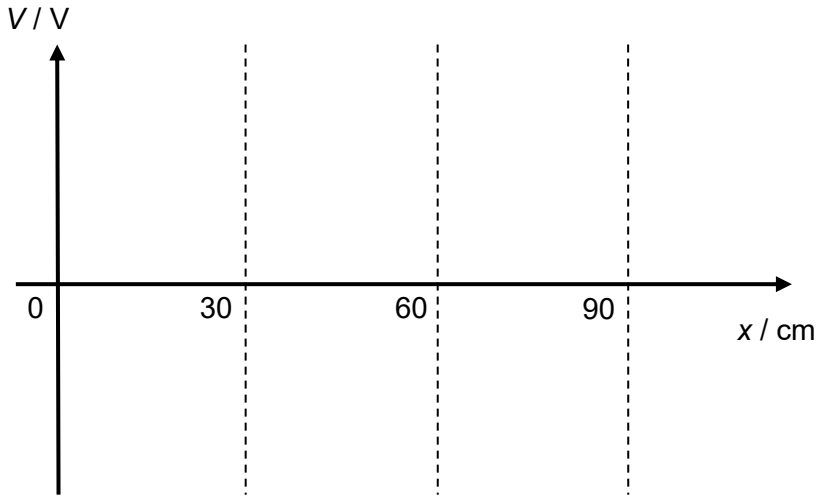

Fig. 8.2

[2]

(ii) On Fig. 8.3, sketch the variation of electric field strength E with distance x between the two plates.

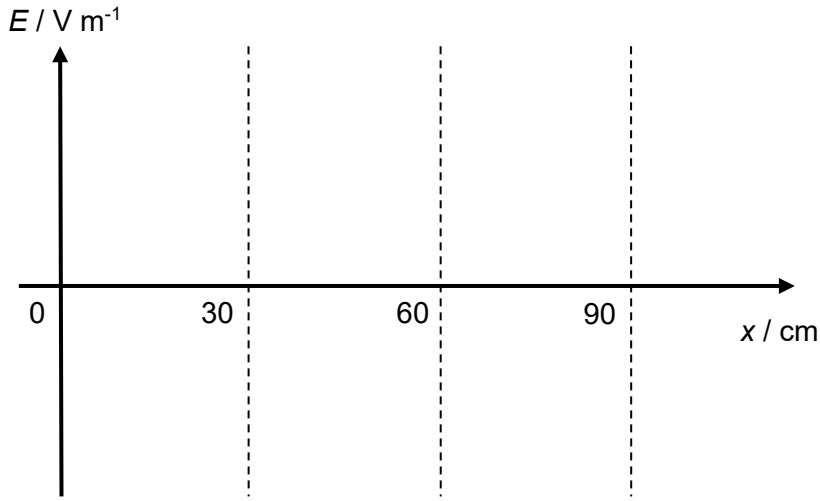

Fig. 8.3

[2]

(c) Fig. 8.4 shows an electron starting from rest between the two vertical plates A and B. The length of the plates is 20.0 cm and they are 90.0 cm apart.

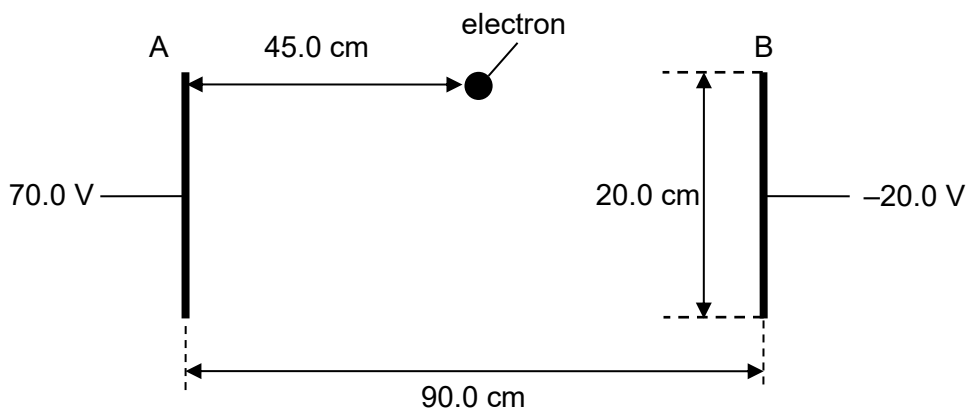

Fig. 8.4

(i) Describe the path travelled by the electron between the two plates A and B,

.....

..... [1]

(ii) Determine the horizontal speed at which the electron would hit one of the plates.

speed = ... m s^{-1} [2]